Security in Large Scale Systems

Week 5 – Project

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(vitorcunha@ua.pt)

Evaluation

Practical Component

Through two projects carried out throughout the semester, with final requirements released ~30 days before.

- Proj1: **November 22** [Threat detection, monitoring, and prevention]
- Proj2: **December 23** [Attack Tolerance Mechanisms]

There will be a bank of hours shared between the two deliveries, totaling 168 hours (i.e., one week)

Project 1 – Threat detection, monitoring and prevention

Objective: Design and run a service-based architecture that scales to large-scale.

- Report: 50%
- Demo/Repositories: 50%

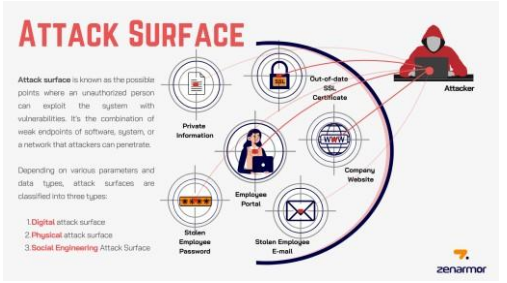
Report

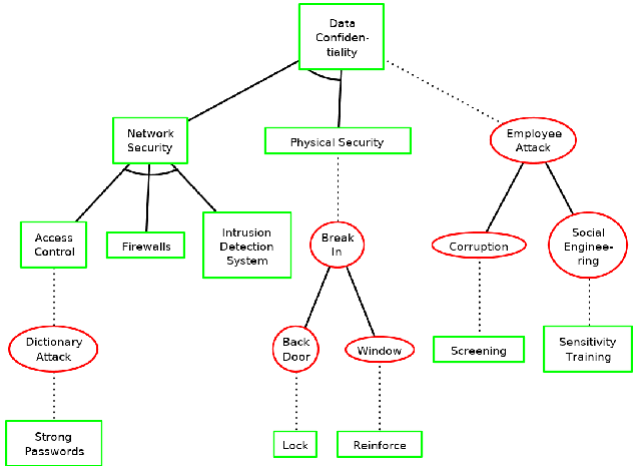
Classic article/thesis structure:

- Introduction
- Relevant State of the Art
- Service Architecture Used
- Risk Management and Mitigation
- Approach to three threats considered in the course syllabus
    - Others may also be considered, confirm with the conductor.
- Evaluation of the solution's approaches and scalability.
- Conclusions

Syllabus (2/4)

2. Risk Management and Mitigation
   - Attack surfaces
   - Attack and Defense Trees
   - Large-scale system monitoring







Syllabus (3/4)

3. Threat detection, monitoring, and prevention **(Project #1 - 22/11)**

- Route and signals hijacking
- Massive denial-of-service disruptions
- Advanced persistent threats
- Botnets
- Junk email
- Fraud and social attacks
- Reputation Systems





Key aspects to consider

- Identification/registration of elements (i.e., name construction) in a scalable way:
  - Where is that element either in physical terms (real machine) or in logical terms (containers / virtual machines).
  - Geographic location and delegation of responsibilities
  - Ability to monitor elements by location, optimizing consumed bandwidth (bonus)
  - Ability to quickly locate event/incident source

Monitoring

- Use Prometheus as in class :)

- Valued ability to generate events for (general) anomalies and subsequent correlation for root cause detection.

Fault Management Workflow (FCAPS)



Fault Management within FCAPS does not deal with security events (i.e., at least explicit security events)

Threats: Integrate into SIEM

- It is recommended to use Wazuh, as in the practical classes.

- Detect incident (**3x types**) and create appropriate action-response.

What is a SIEM? *(Focus on the first)*

It is a **Security Information and Event Manager**, consisting of:

- Log management
- Security Information Management (SIM)
- Security Event Manager (SEM)
- Normalization and Correlation
- Managed Security Service (MSS)
- Compliance and Reporting

Discussion of results

- Always remember to explain:
  - The description of the threat
  - Why is it relevant to the system?
  - How can it be exploited?
  - What are the relevant monitoring/mitigation actions/metrics?
  - What are the action-responses adopted?
  - Post-mortem of PoC(s) *(i.e., practical validation of the solution)*

Conclusions

- List the main findings and contributions:
  - Describing why the solution is not perfect also earns points (i.e., this is knowledge)
  - To make a bridge for future work.

Demo

- In class, at a later date (to be communicated to each student)

Doubts and fine-tuning of criteria